

Toward Sufficient Access: Measuring Accessibility Poverty Through Perceived Accessibility

Md Hamidur Rahman, Alex Karner

Graduate Program in Community and Regional Planning



ACSP 65th Annual Conference
Oct 23-25 | Minneapolis

A Story : Moving from Disparity to insufficiency

Person A

- Female, African American, and has a disability
- Resides in a downtown neighborhood
- Commutes primarily by public transit
- Has good access to amenities near home

Person B

- Male, White, and without a disability
- Resides in a suburban area
- Commutes primarily by private car
- Has limited access to nearby amenities

who has better access ?

vs

who has sufficient access?

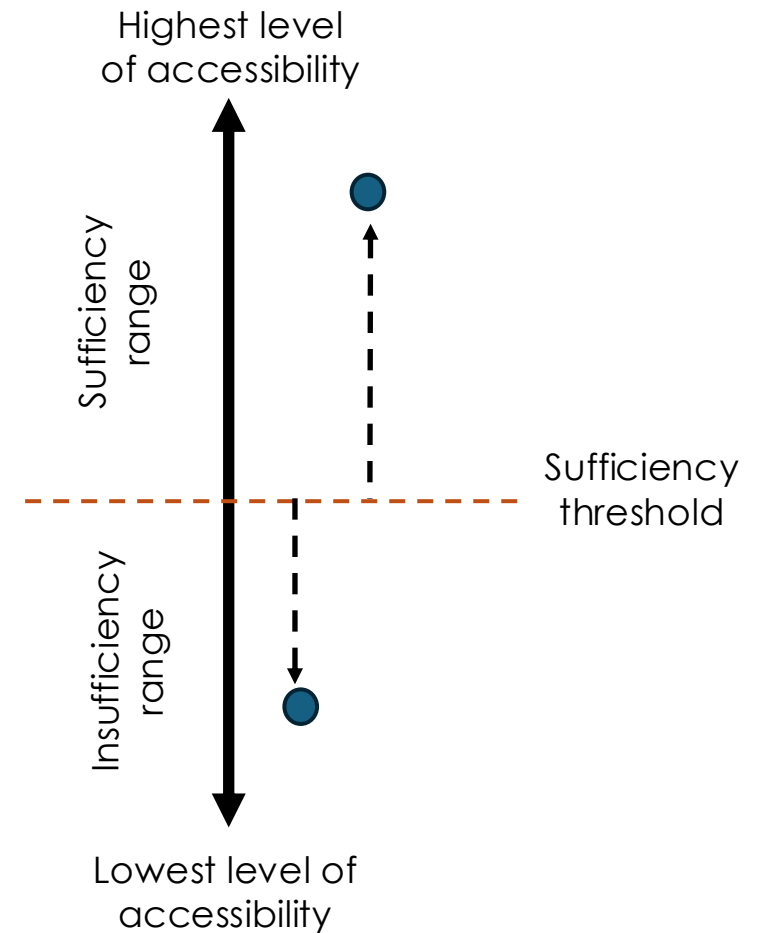
Disparity



Insufficiency

Accessibility Poverty: A Sufficiency Argument

- **Accessibility poverty:** a state in which a person's ability to access essential activities including work, school, health care, and leisure becomes so low that they are unable to satisfy basic needs deemed normal in a particular society (Lucas, 2012; Martens & Bastiaanssen, 2019; Karner et al., 2024)



Source: Adapted from Martens, K., 2017. Transport Justice: Designing Fair Transportation Systems. Routledge, New

Empirical Study on Sufficient Accessibility

- Only a handful of research delved into setting a possible sufficiency threshold.
- Most of the work compared transit accessibility with auto-accessibility for threshold determination. (Martens, 2016; van der Veen et al., 2020 ; Martens et al., 2022; Karner et al., 2024)
- Sufficiency criteria: transit access is sufficient if it provides X % of auto access.

Access to jobs in 30 minutes of auto	Access to jobs in 30 minutes of transit	Transit is providing 20% of jobs that are available with auto within 30 minutes
50,000 jobs	10,000 jobs	

Transit access as sufficient for a place (i.e., block groups or a census tracts) → if it provides 10%, 20%, 30%, 40%, or 50% of automobile access.

New Direction on Understanding Sufficient Access

Shift in focus: (1) objective conditions of accessibility to the lived travel experiences

(Singer & Martens, 2023; De Vos, 2024).

Goal: Understanding individual's **perception of access**



Travel Problems



Ease of Travel

New Direction on Understanding Sufficient Access

Shift in focus: (2) Linking of accessibility to travel or life outcomes.

(Lättman et al., 2019; Allen & Farber, 2020; Smart & Klein, 2020; Ettema et al., 2025; Pot et al., 2025; Palm et al., 2025).

Goal: understanding **causal consequences** of accessibility improvements i.e., introducing a new BRT line a city or shared bike lane in a neighborhood.

Our study focus: Empirically linking spatial and perceived accessibility to activity participation and life satisfaction.

Goal: offering a way to identify *sufficient accessibility thresholds*

Research Questions

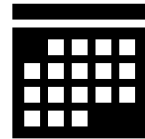
- ✓ **RQ 1:** What is the nature of the relationship between **accessibility and activity participation**?
- ✓ **RQ 2:** How do **calculated (spatial) accessibility and perceived accessibility** jointly shape people's ability to achieve satisfactory levels **of well-being**?
- ✓ **RQ 3:** Is there evidence of a **threshold of sufficient access**, beyond which greater accessibility no longer improves participation or satisfaction?

Data and Methods

Data



Online Survey



Jul 15-Aug 15 (2025)



Total 670 Sample
*616 Final Sample



Qualtrics Panel on
US General Population



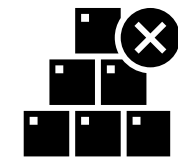
54 Questions



completion time ~ 12 mins

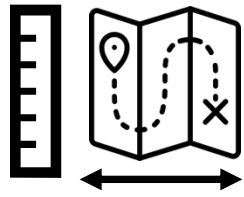


Quotas:
Gender, Age, Race,
Ethnicity and Region



2 attention check
Completion time
Straight-lining

Key Variables



Spatial accessibility

- Jobs within 45 minutes auto travel time
- Jobs within 45 minutes of transit commute
- Walkability index score

Source: EPA Smart Location Database V3



Perceived accessibility

- A 4 item PAC scale instrument of Lättman et al. (2016, 2018)
- Each item rated on Likert-type scale (1-5) from Strongly disagree to strongly agree



Activity Participation

- Number of unique destinations visited outside of home during last week
- Unmet Activity Needs (Binary: Yes or NO)

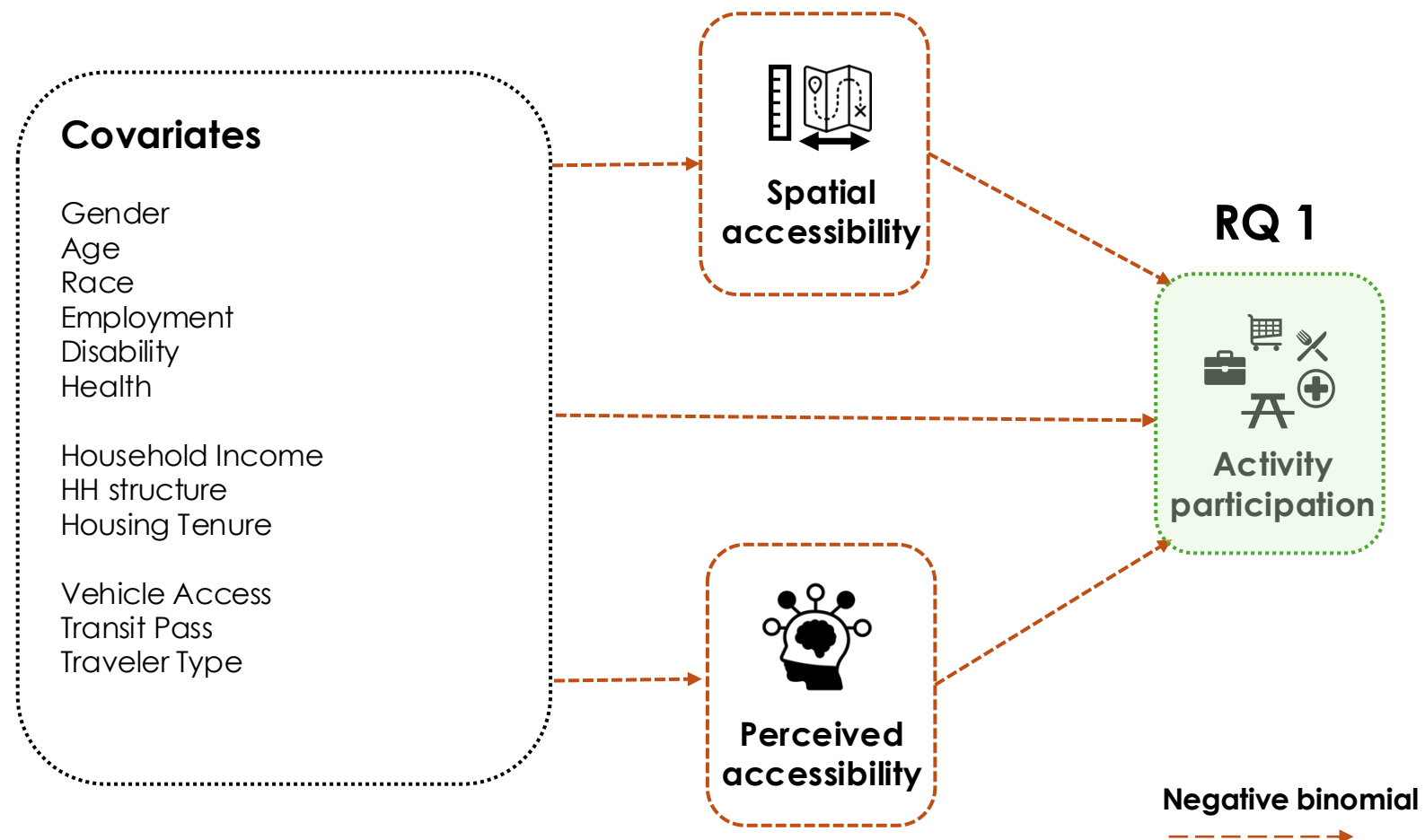


Life Satisfaction

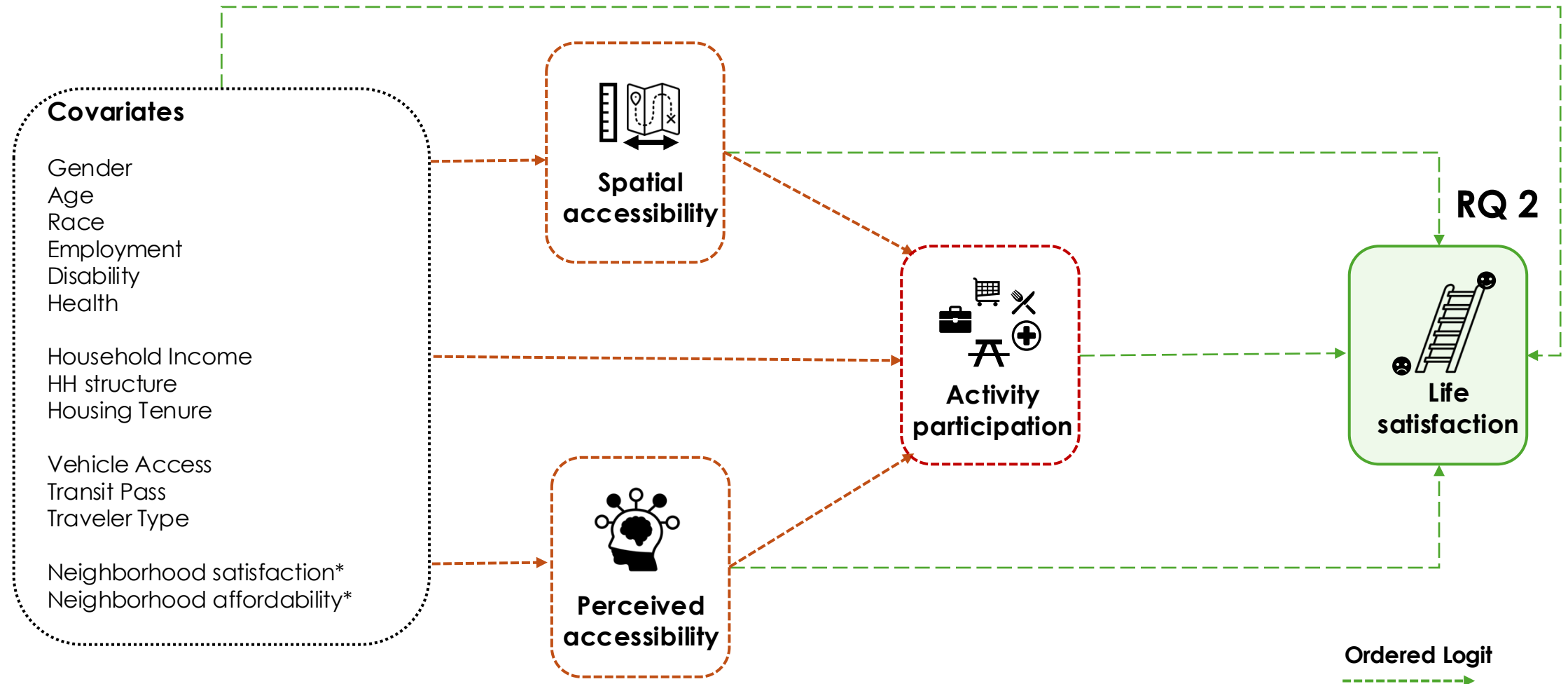
- Cantril Self-Anchoring Ladder with steps from 0 (worst possible life) to 10 (best possible life)
- A single-item instrument
- Widely used in well-being and quality of life research

Source: Online panel data

Analytical Framework



Analytical Framework



Results

RQ 1: Negative Binomial Model

Dependent variable: Number of unique activities in the past week

Observations= 584
 R^2 Nagelkerke (0.154)
 AIC (2604.873)
 2 x log-likelihood (-2560.873)



Lower activity participation (IRR<1)

Age (mean centered) (0.994)

Poor Health (ref. Excellent health) (0.950)

Not-working (ref. Employed) (0.880)

Have Disability (ref. No) (0.950)

Higher Composite Spatial Accessibility (0.999)

Male (ref. Female) (0.93)

White (ref. Non-white) (0.997)

Student (ref. Employed) (0.936)

HH Income [\$40k-\$75k] (ref. <\$40k/yr) (0.936)



Higher activity participation (IRR>1)

Higher Perceived Accessibility (1.134)

Good Health (ref. Excellent health) (1.074)

Vehicle deficit HH (ref. Zero car HH) (1.191)

1+ Vehicles per Driver HH (ref. Zero car HH) (1.281)

Transit Pass- Yes (Ref. No) (1.153)

Renter (ref. Owner) (1.103)

HH Income [75-120k/yr] (ref. <40k/yr) (1.056)

HH Income [120k+/yr] (ref. <40k/yr) (1.025)

Living Alone-Yes (ref. No) (1.036)

RQ 2: Ordered Logit Model Result

Dependent variable: Life satisfaction (ordered as low-med-high)

Observations= 584
R² Nagelkerke (0.337)
AIC (1127.385)
2 x log-likelihood (-535.69)



Low life satisfaction (IRR<1)

Poor Health (ref. Excellent health) (0.347)

Living alone (ref. No) (0.596)

Vehicle deficit (ref. Zero-car HH) (0.495)

1+ Vehicles per Driver HH (ref. Zero-car HH) (0.539)

Higher Composite Spatial Accessibility (0.985)

Higher Activity participation (0.989)

Female (ref. Male) (0.835)

Have disability (ref. No) (0.781)

Unaffordable neighborhood (ref. Neutral) (0.990)

Unsatisfied with neighborhood (ref. Neutral) (0.861)



Higher life satisfaction (IRR>1)

Higher Perceived Accessibility (2.205)

Satisfied with neighborhood(ref. Neutral) (1.788)

Age (mean centered) (1.022)

Trips last day (1.042)

White (ref. Non-white) (1.155)

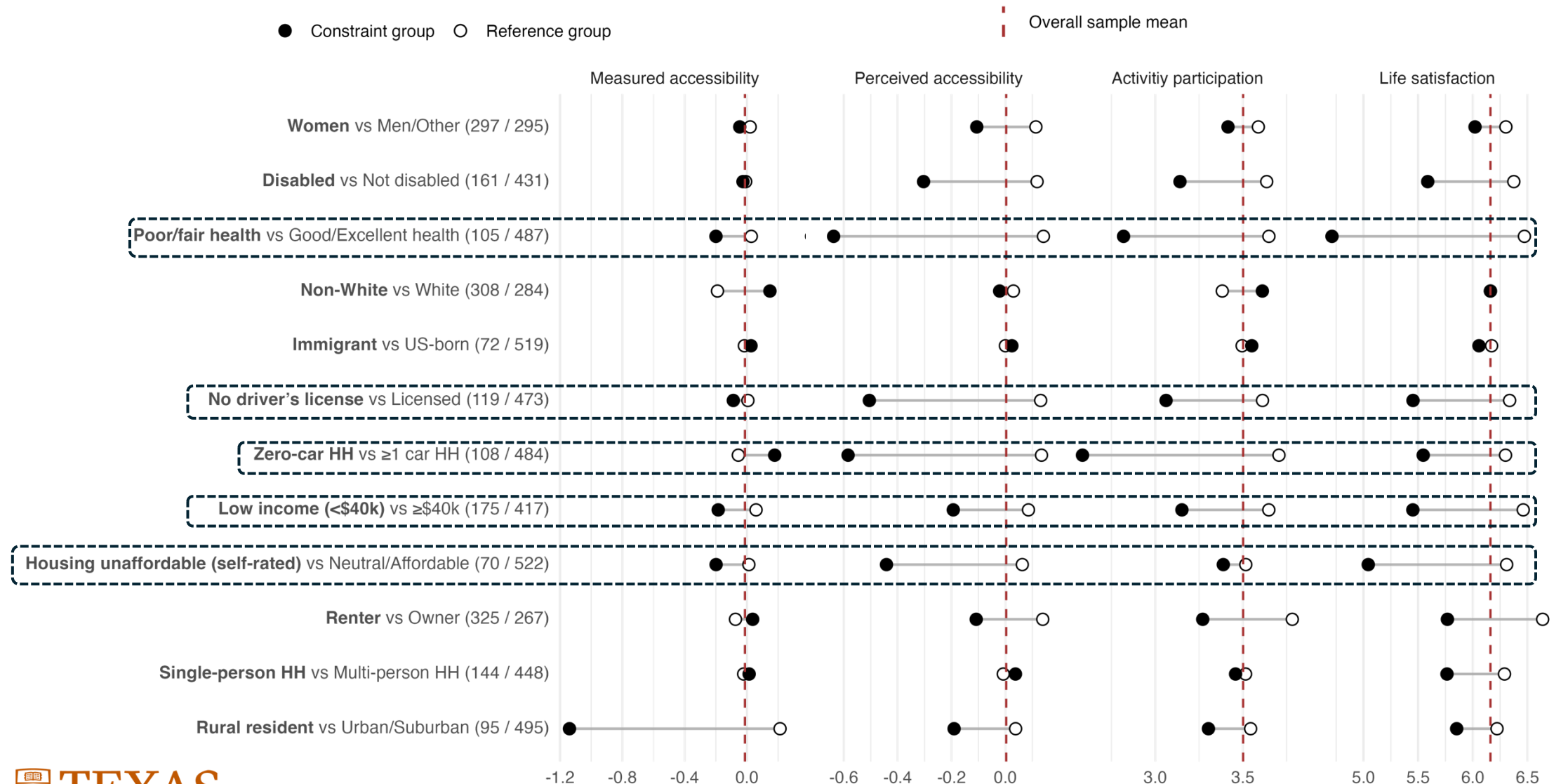
Not working/Student (ref. Employed) (1.184)

Renter (ref. Owner)

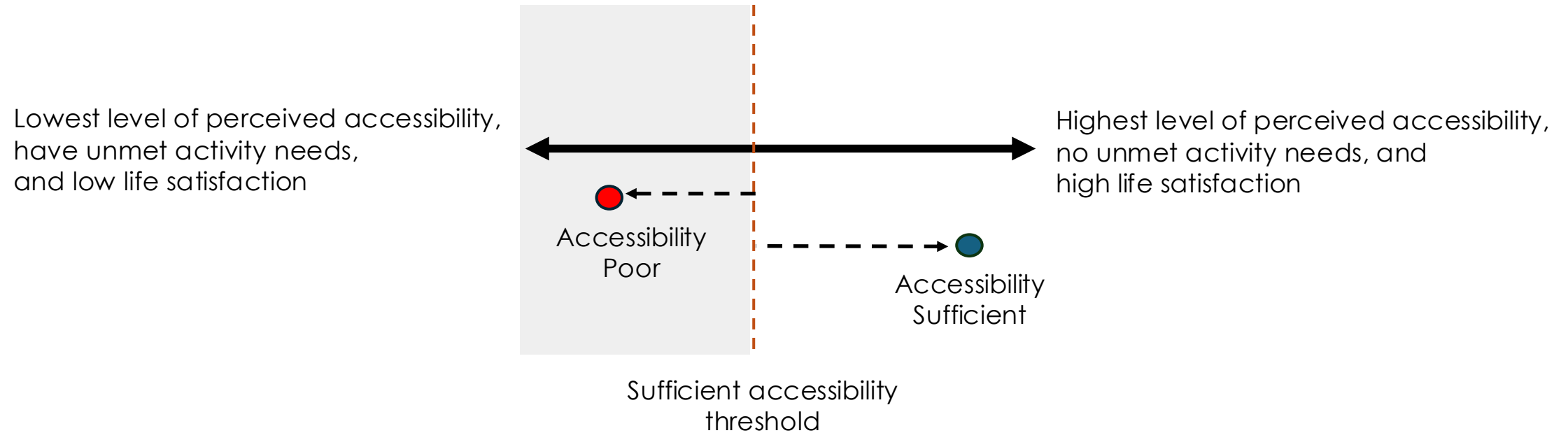
Medium/high HH Income (ref. HH Income [<\$40k/yr]

Affordable neighborhood (ref. Neutral)

RQ3: Determining Sufficient Accessibility



RQ3: Determining Sufficient Accessibility



priority can be given to those who suffer the largest insufficiencies.

Accessibility Poverty

Perceived Accessibility Poor (PA<0) And Low Activity Participation (<3 activities)	Perceived Accessibility Poor (PA <0) And Have Unmet Needs	Perceived Accessibility Poor (PA<0) And Low Life Satisfaction (<6.14)
126 (21.3 %)	157 (26.5%)	177 (29.9%)

Key Findings



Perceived Accessibility Matters Most.

Policy Implications

- Incorporating perceived accessibility in **tradition travel or activity-based survey** to better capture lived experiences transportation needs and barriers.
- Standard setting for accessibility needs **meaningful public engagement**, really giving people to shape their preference since people people would decide what level of access is reasonable or sufficient for their needs.
- Explicitly **prioritizing groups** of people, namely those who are **suffering from the largest insufficiencies**.

Reference

- Allen, J., & Farber, S. (2020). Planning transport for social inclusion: An accessibility-activity participation approach. *Transportation Research Part D: Transport and Environment*, 78, 102212. <https://doi.org/10.1016/j.trd.2019.102212>
- Ettema, D., van Lierop, D., Fu, X., van den Berg, P., & Geigenmüller, I. (2025). What is Transport Adequacy? Quantifying experienced transport poverty in the Netherlands. *Travel Behaviour and Society*, 39, 100982.
- Karner, A., Pereira, R. H. M., & Farber, S. (2024). Advances and pitfalls in measuring transportation equity. *Transportation*. <https://doi.org/10.1007/s11116-023-10460-7>
- Lucas, K. (2012). Transport and social exclusion: Where are we now?. *Transport policy*, 20, 105-113.
- Lättman, K., Olsson, L. E., Friman, M., & Fujii, S. (2019). Perceived accessibility, satisfaction with daily travel, and life satisfaction among the elderly. *International journal of environmental research and public health*, 16(22), 4498.
- Martens, K. (2016). *Transport justice: Designing fair transportation systems*. Routledge.
- Smart, M. J., & Klein, N. J. (2020). Disentangling the role of cars and transit in employment and labor earnings. *Transportation*, 47 (3), 1275–1309. <https://doi.org/10.1007/s11116-018-9959-3>
- Pot, F. J., Heinen, E., & Tillemans, T. (2025). Sufficient access? Activity participation, perceived accessibility and transport-related social exclusion across spatial contexts. *Transportation*, 52(4), 1679-1707.
- Palm, M., Raynor, K., & Grajdura, S. (2025). Bridging access with life satisfaction and Nussbaum's core capabilities. *Transportation Research Part D: Transport and Environment*, 146, 104877.
- van der Veen, A. S., Annema, J. A., Martens, K., Van Arem, B., & de Almeida Correia, G. H. (2020). Operationalizing an indicator of sufficient accessibility—a case study for the city of Rotterdam. *Case studies on transport policy*, 8(4), 1360-1370.

Thank You!



Md Hamidur Rahman

PhD Student in Community and Regional Planning

hamidur.rahman@utexas.edu